

Practical 8

Spanning Tree Protocol (STP) Simulation Guide

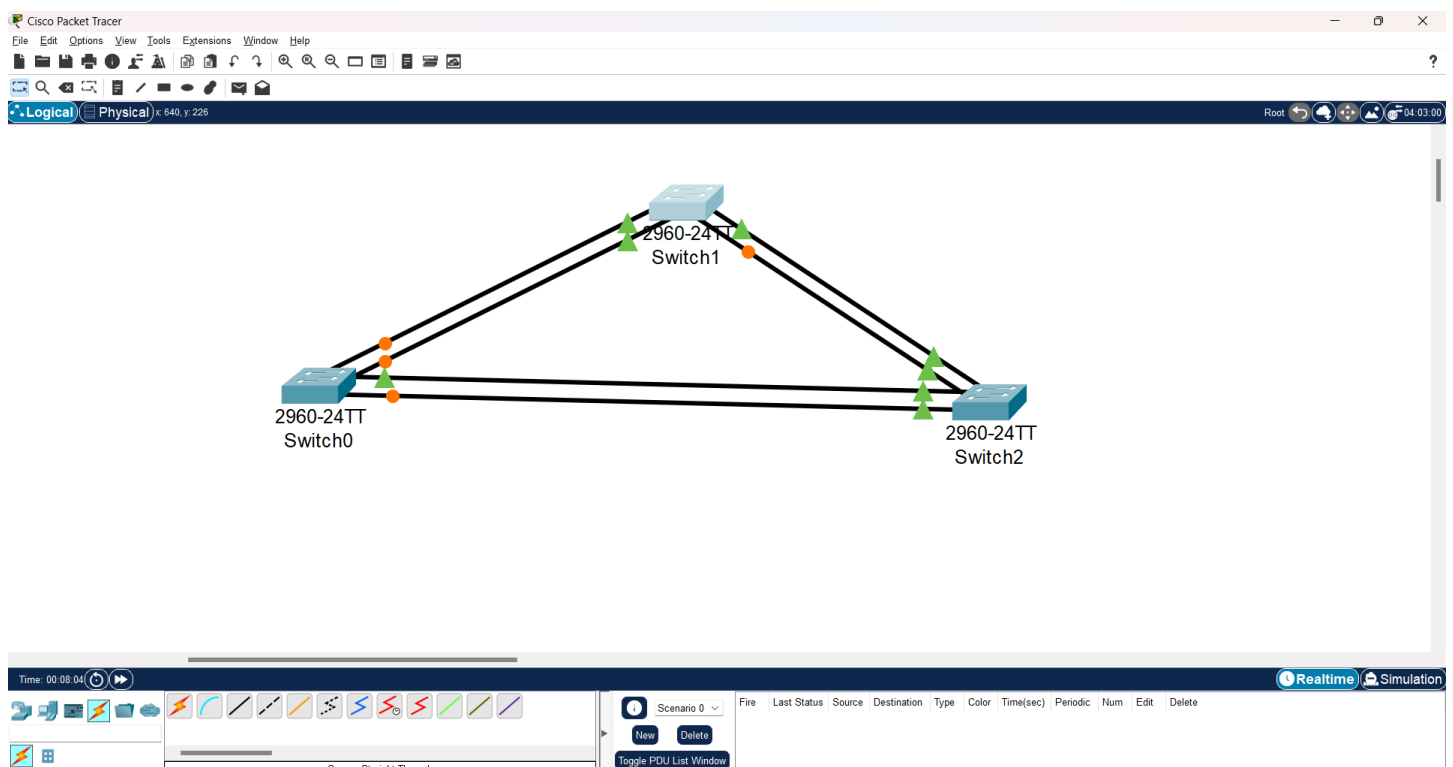
Objective:

Prevent network loops using Spanning Tree Protocol (STP) to ensure network stability.

Steps:

1. Create a Network with Redundant Links

- **Open Cisco Packet Tracer.**
- **Add at least three switches** to the workspace (e.g., Switch0, Switch1, and Switch2).
- **Interconnect the switches with redundant links.** For instance:
 - Connect **Switch0** to **Switch1** with two links.
 - Connect **Switch1** to **Switch2** with two links.
 - Connect **Switch0** to **Switch2** with two links.
- Use **crossover cables** for inter-switch connections if using older switches; otherwise, use straight-through cables.



2. Enable STP on the Switches

- **Configure the switches:**
 1. Go to the CLI (Command Line Interface) of each switch (repeat the same step for each switch).

2. Enter the following commands:

Code:

```
enable
configure terminal
spanning-tree vlan 1
exit
```

```
Switch>enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#spanning-tree vlan 1
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

- The spanning-tree vlan 1 command enables STP for VLAN 1 (default VLAN).

• **Verify STP operation:**

- Use the show spanning-tree command to observe the status of STP. The output should show:
 - The **Root Bridge** (the switch elected as the root).
 - The **ports in forwarding state**.
 - The **ports in blocking state** (redundant links).

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.C90E.99A5
             Cost        19
             Port        3(FastEthernet0/3)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     000A.41AE.17B1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/4	Altn	BLK	19	128.4	P2p
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/3	Root	FWD	19	128.3	P2p

```
Switch#
```

3. **Observe STP in Action**

- Once STP is enabled:
 - Check which ports are **blocked** to prevent loops.

- The show spanning-tree command will display which interfaces are in **Forwarding** or **Blocking** states.

4. Simulate a Link Failure

- Disconnect one of the **primary links** between two switches. For example:
 - Disconnect a link between Switch0 and Switch1.
- Observe how STP reacts:
 - Use the show spanning-tree command to verify that a **blocked link** is now in the **forwarding state**, restoring network connectivity.

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.C90E.99A5
             Cost        19
             Port        3(FastEthernet0/3)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     000A.41AE.17B1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/4	Altn	FWD	19	128.4	P2p
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/3	Root	FWD	19	128.3	P2p

```
Switch#
```

5. Verify Network Stability

- Use **ping** to ensure connectivity between devices on the network remains intact during the simulation.
- Verify that STP successfully prevented any loops.

Outcome:

By completing this simulation, you will:

- Understand how STP blocks redundant links to prevent network loops.
- See STP's role in automatically reactivating links to maintain network stability when a primary link fails.
- Gain practical knowledge of configuring and monitoring STP in a network environment.